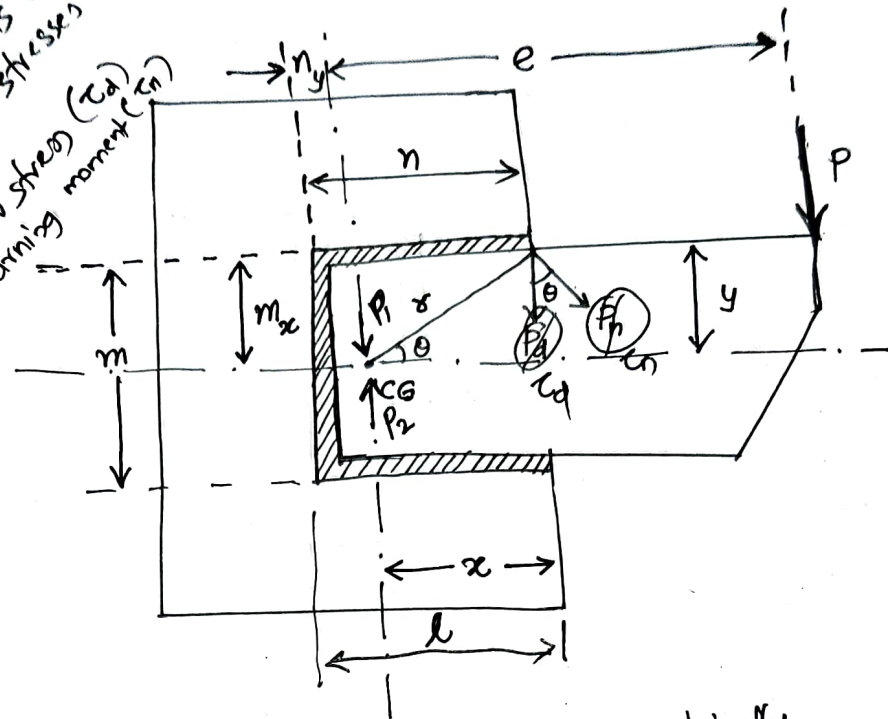


Eccentrically Loaded Weld

When a welded joint is loaded eccentrically, 2 types of stresses are induced:

- ① Direct / primary shear stress (τ_d)
- ② Shear stress due to turning moment (τ_m)



$P \rightarrow$ Load acting on the weld. eccentrically.

$n \rightarrow$ Horizontal length of the weld (in mm)

$m \rightarrow$ Vertical length of the weld. (in mm)

$CG \rightarrow$ Centre of gravity of weld.

$m_x \rightarrow$ Distance of CG from top.

$n_y \rightarrow$ Distance of CG from the left.

$x \rightarrow$ Distance of CG from right side top corner.

$e \rightarrow$ Distance of CG to the load. (eccentricity)

$P_d \rightarrow$ Direct load per unit length of the weld (N/mm)

$P_n \rightarrow$ Normal load per unit length of weld (N/mm)

$\theta \rightarrow$ Angle b/w P_d & P_n .

$P_r \rightarrow$ Resultant load per unit length of weld.

$h/h_t \rightarrow$ Throat thickness.

$$h_t = 0.707h / 0.707w \cdot 0.7075.$$

$h/s/w \rightarrow$ thickness/size of weld.

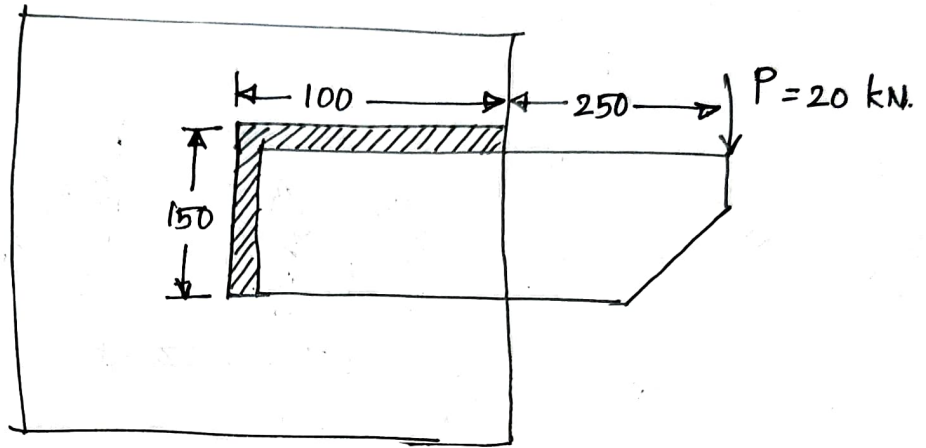
$\tau \rightarrow$ Allowable shear stress in weld

$l \rightarrow$ ~~Total~~ length of weld.

- ① Determine the size of the weld for plate welded as shown in figure. The allowable stress for weld 75 MPa. Direct Load 80 N/mm?

Given

$$\begin{aligned}m &= 150 \text{ mm} \\n &= 100 \text{ mm} \\P &= 20 \text{ kN} \\T &= 75 \text{ MPa} \\P_d &= 80 \text{ N/mm}\end{aligned}$$



Step-1

Find C.G. [page 96].

$$m_x = \frac{m^2}{2(n+m)} = \frac{150^2}{2(100+150)} = 45 \text{ mm}$$

$$n_y = \frac{n^2}{2(n+m)} = \frac{100^2}{2(100+150)} = 20 \text{ mm}$$

Step-2

Find $r, e, \cos \theta$.

$$r = \sqrt{x^2 + y^2}$$

$$\begin{aligned}x &= n - n_y \\&= 100 - 20 = \underline{80 \text{ mm}}\end{aligned}$$

$$y = m_x = \underline{45 \text{ mm}}$$

$$r = \sqrt{80^2 + 45^2} = \underline{91.78 \text{ mm}}$$

$$e = x + 250$$

$$= 80 + 250 = \underline{\underline{330 \text{ mm}}}$$

$$\cos \theta = \frac{x}{r} = \frac{80}{91.78}$$

$$\therefore \theta = \cos^{-1}\left(\frac{80}{91.78}\right) = \underline{\underline{29.34^\circ}}$$

$$\text{Length of weld} = 150 + 100 = \underline{\underline{250 \text{ mm}}} \quad (n+m)$$

Step-3

Direct load per unit length of weld

$$P_d = \frac{P}{l} = 80 \text{ N/mm (given)}$$

Direct stress, $\tau_d = \frac{P_d}{s} = \underline{\underline{\frac{80}{s}}}$

Normal stress, $\tau_n = \frac{P \times e \times r}{J}$

$J \rightarrow$ polar moment of inertia (page 96, Table 6.7)

$$J = h_t \times \frac{(n+m)^4 - 6n^2m^2}{12(n+m)}$$

We know, $h_t = 0.707 \times s$ (thickness of throat).

$$\therefore J = \frac{0.707 s \times (100+150)^4 - 6 \times 100^2 \times 150^2}{12(100+150)}$$

$$= \underline{\underline{602422.91 \times s}}$$

$$\therefore \tau_n = \frac{P \times e \times r}{J} = \frac{2 \times 10^3 \times 330 \times 91.78}{602422.91 \times s} = \underline{\underline{\frac{1005.51}{s}}}$$

Step-4

(Resultant) or allowable shear stress

$$\tau = \sqrt{\tau_d^2 + \tau_n^2 + 2\tau_d\tau_n\cos\theta} = \sqrt{\left(\frac{80}{s}\right)^2 + \left(\frac{1005.51}{s}\right)^2 + 2 \times \frac{80 \times 1005.51}{s^2} \times \cos 29.34}$$

$$75 = \frac{6400}{s^2} + \frac{1011050.36}{s^2} + \frac{140244.89}{s^2} \Rightarrow \therefore s^2 = \frac{1157695.25}{75^2}$$

Throat Thickness, $h_t = 0.707 \times s$
 $= 0.707 \times 14.34$
 $= \underline{\underline{12.06 \text{ mm}}}$

Thickness of weld, $\therefore s = \underline{\underline{14.34 \text{ mm}}}$